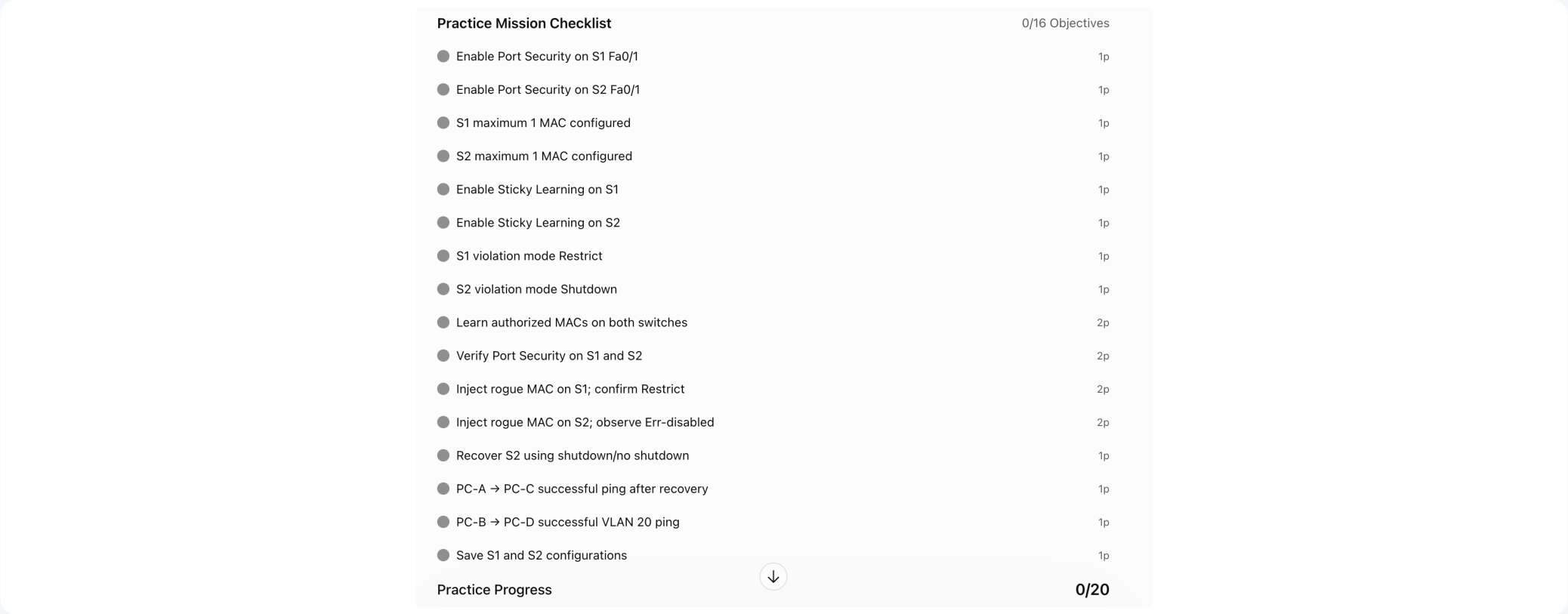
Switch Port Security

Personal learning project | Practical 01 of 07 | Lesson 21

Practised securing S1/S2 access ports with a maximum of one secure MAC address and sticky learning. Explored restrict versus shutdown violations, verification and recovery from an err-disabled interface.



Lab scope: port-security configuration, authorized and rogue MAC testing, verification, recovery, connectivity checks and saved switch configurations.

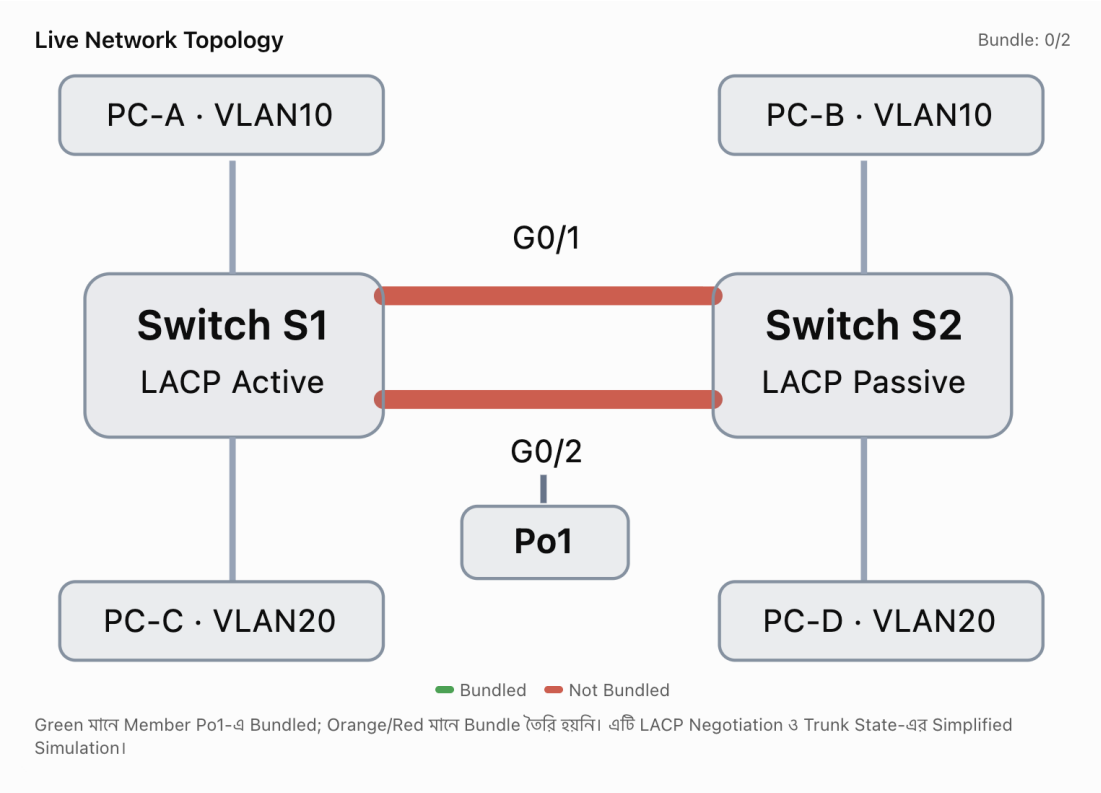
Secure MAC limits | Sticky learning | Violation modes | Interface recovery

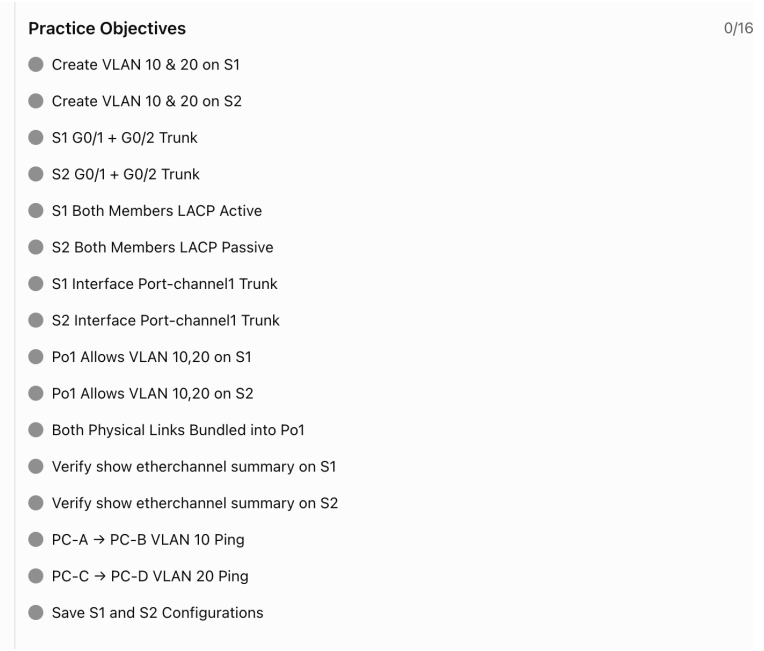
Guided simulator practice. Images document lab scenarios and objectives, including initial fault states.

EtherChannel with LACP

Personal learning project | Practical 02 of 07 | Lesson 20

Practised bundling G0/1 and G0/2 into Port-channel 1 using LACP active/passive modes. Covered VLAN 10/20 trunks, allowed VLANs, bundle verification and same-VLAN ping tests.





Two-switch, four-PC scenario. The captured view shows links before a successful bundle is formed.

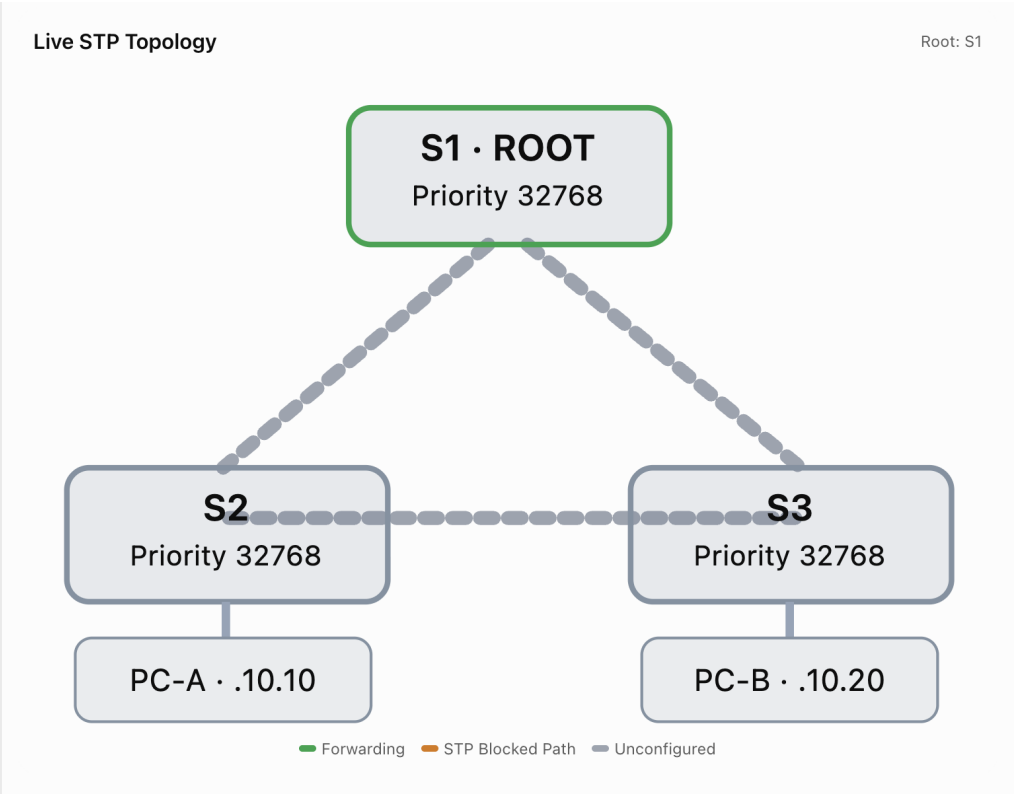
LACP | Port-channel | Trunk consistency | Bundle verification

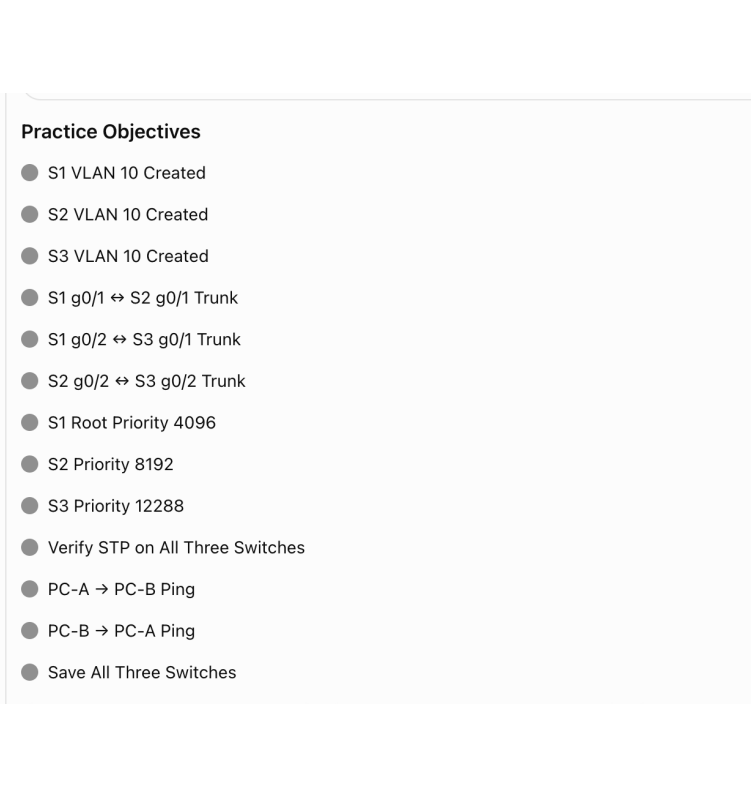
Guided simulator practice. Images document lab scenarios and objectives, including initial fault states.

Spanning Tree Protocol

Personal learning project | Practical 03 of 07 | Lesson 19

Practised VLAN 10 and trunk configuration across three switches in a triangle topology. Covered root-bridge priorities, STP verification, redundant paths and PC-to-PC connectivity.





Three-switch STP scenario. The capture shows the starting priorities and unconfigured paths; the checklist defines the target configuration.

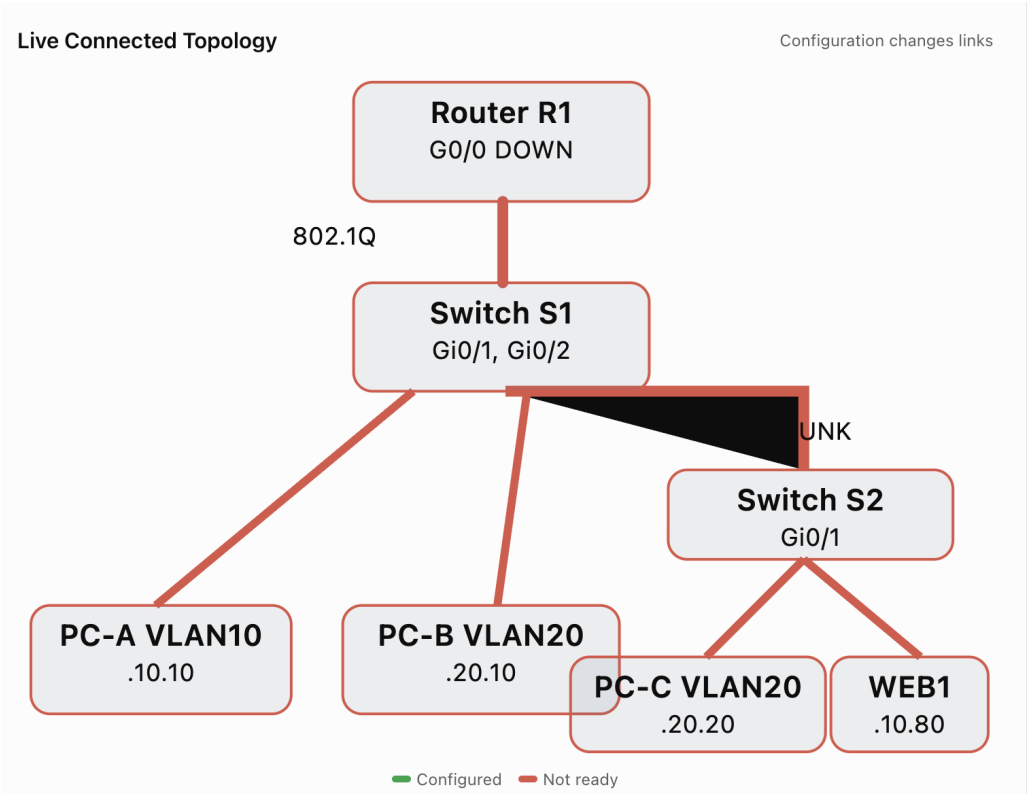
Root bridge | Bridge priority | Redundant switching | Loop prevention

Guided simulator practice. Images document lab scenarios and objectives, including initial fault states.

VLANs and Inter-VLAN Routing

Personal learning project | Practical 04 of 07 | Lesson 18

Practised VLAN 10/20 access ports, switch-to-switch trunks and router-on-a-stick routing. Covered 802.1Q subinterfaces, VLAN gateways and same-VLAN/inter-VLAN ping checks.



Practice Objectives

- Create VLAN 10 & 20 on S1
- Create VLAN 10 & 20 on S2
- S1 Fa0/1 — Access VLAN 10
- S1 Fa0/2 — Access VLAN 20
- S2 Fa0/1 — Access VLAN 20
- S2 Fa0/2 — Access VLAN 10
- S1 G0/1 — Trunk to R1
- S1 G0/2 & S2 G0/1 — Trunks
- R1 G0/0 Physical Interface UP
- R1 G0/0.10 — 802.1Q VLAN 10 + Gateway
- R1 G0/0.20 — 802.1Q VLAN 20 + Gateway
- PC-A → PC-B Inter-VLAN Ping
- PC-B → PC-C Same-VLAN Trunk Ping
- PC-C → WEB1 Inter-VLAN Ping
- Save R1, S1 & S2 Configurations

Router, two switches, PCs and a web host. The captured scenario includes the initial router-interface-down fault.

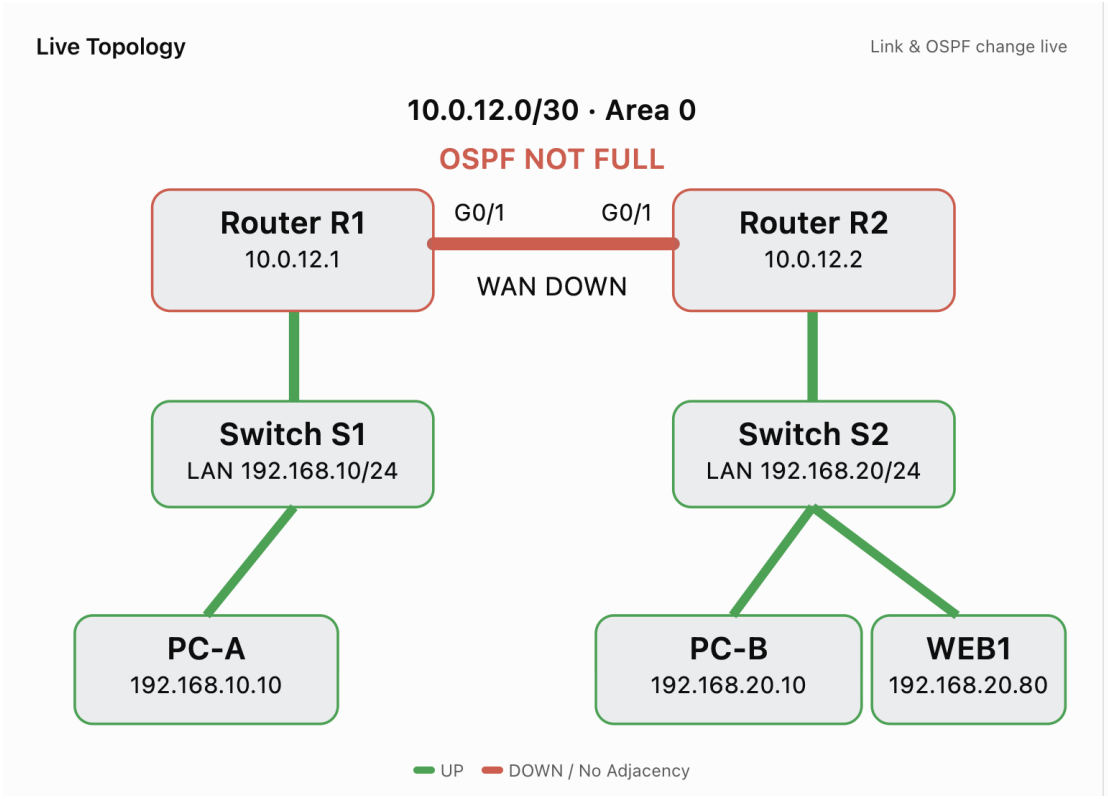
VLAN segmentation | Access ports | 802.1Q trunks | Router-on-a-stick

Guided simulator practice. Images document lab scenarios and objectives, including initial fault states.

Single-Area OSPF

Personal learning project | Practical 05 of 07 | Lesson 17

Practised OSPF Area 0 between two routers over 10.0.12.0/30. Covered WAN interfaces, LAN advertisements, FULL neighbor verification, learned routes and bidirectional connectivity checks.



Practice Objectives

- R1 WAN G0/1 UP
- R2 WAN G0/1 UP
- R1 OSPF Area 0 WAN
- R2 OSPF Area 0 WAN
- Advertise Meidling LAN 10
- Advertise Favoriten LAN 20
- Verify FULL Neighbor from Both Routers
- Verify Learned OSPF Routes on Both
- PC-A → PC-B Ping
- PC-B → PC-A Ping
- Save R1 and R2 Configurations

Two-site routing scenario connecting LAN 10 and LAN 20. The captured WAN is down and OSPF is not yet FULL.

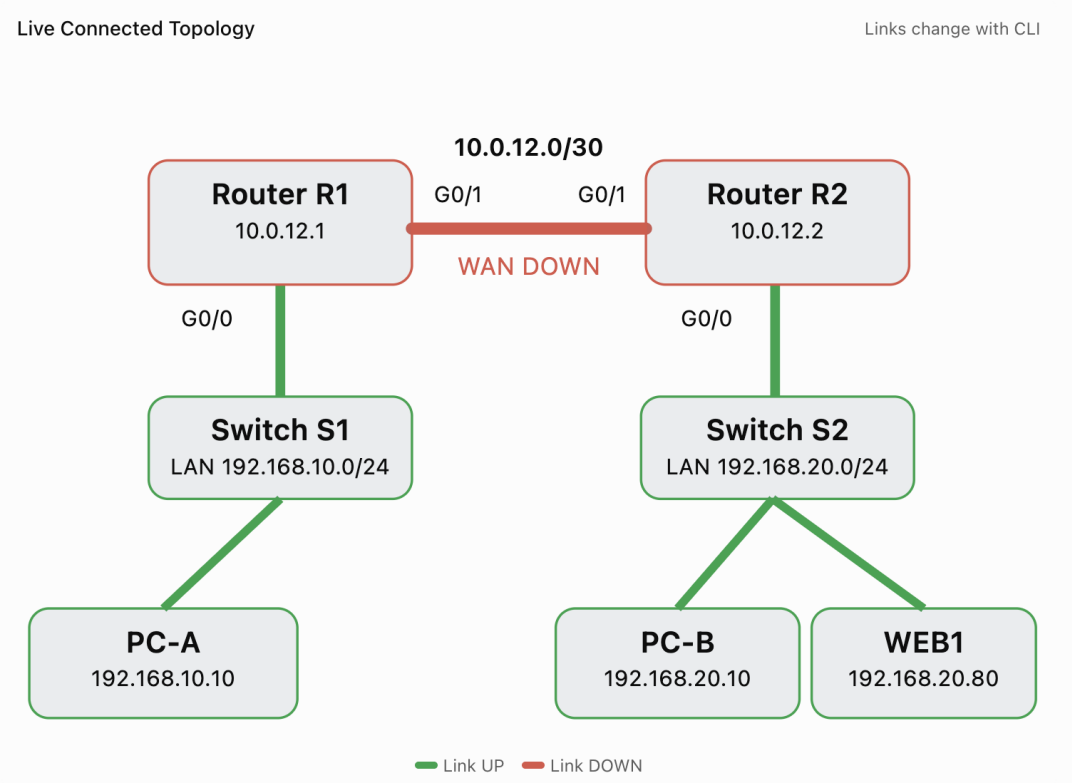
OSPF Area 0 | Adjacencies | Route advertisements | Routing verification

Guided simulator practice. Images document lab scenarios and objectives, including initial fault states.

Static and Default Routing

Personal learning project | Practical 06 of 07 | Lesson 16

Practised restoring a two-router WAN and configuring a static route on R1 with a default route on R2. Covered routing-table checks, PC/web-server ping tests and saving configurations.



Practice Progress

- R1 G0/1 WAN Interface UP
- R2 G0/1 WAN Interface UP
- R1 Static Route to LAN 20
- R2 Default Route via R1
- Verify both Routing Tables
- PC-A → PC-B Ping Successful
- PC-B → PC-A Ping Successful
- PC-A → Web Server Ping Successful
- Save both Router Configurations

Two-router scenario with separate /24 LANs and a /30 WAN. The captured fault state has the WAN link down.

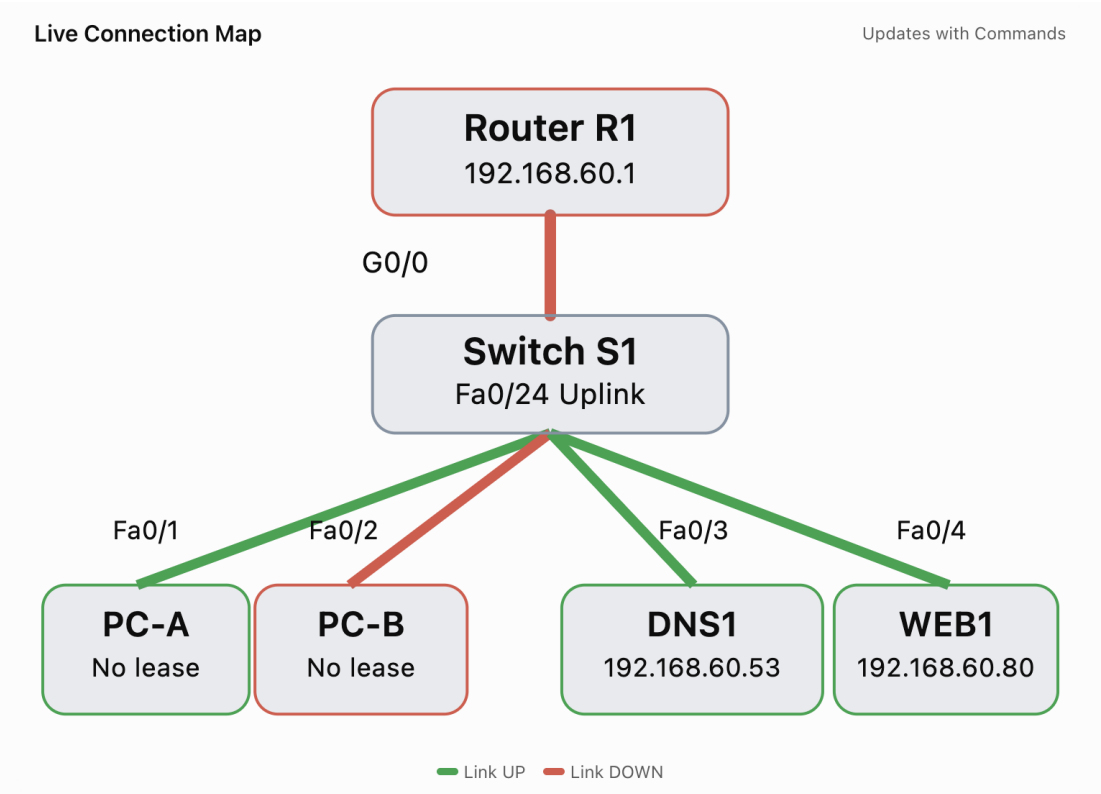
Static routes | Default routes | WAN troubleshooting | End-to-end testing

Guided simulator practice. Images document lab scenarios and objectives, including initial fault states.

DNS and DHCP Troubleshooting

Personal learning project | Practical 07 of 07 | Lesson 15

Practised diagnosing router/switch link faults, DHCP DNS options, the router resolver and DNS A records. Covered client leases, hostname lookup, ping by name and configuration persistence.



Practical Exam Objectives

1. Router G0/0 Link UP
2. Switch Fa0/2 Link UP
3. Correct DHCP DNS Option
4. Router Own DNS Resolver
5. Correct DNS A Record
6. PC-A DHCP Lease
7. PC-B DHCP Lease
8. Both PCs Resolve DNS Correctly
9. Both PCs Ping Website by Name
10. Router Running-config Saved

Router-switch-client-server scenario. The capture shows link faults and missing client leases before remediation.

DNS resolver | A records | DHCP DNS options | Client troubleshooting

Guided simulator practice. Images document lab scenarios and objectives, including initial fault states.